NAME : .....

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN) — path to success

RACE
Regular Analysis through Continuous Exercise

JEE (Main + Advanced) 2023
ENTHUSIAST + LEADER COURSE

SCORE ADVANCED | **MATHEMATICS** | **M.M. 78**

REVISION ASSINGMENT # 19 | (LOG, MOD, TRI AND TE) | TIME : 70 MIN.

SECTION–I

Straight Objective Type (3 Marks each, –1 for wrong answer)

1. $\tan200^\circ(\cot10^\circ - \tan10^\circ) + 1$ is equal to -

(A) $\sqrt{3}$ (B) $\frac{1}{\sqrt{3}}$ (C) 1 (D) 3

2. If $\frac{(\cos1^\circ+\sin1^\circ)(\cos2^\circ+\sin2^\circ)(\cos3^\circ+\sin3^\circ)....(\cos45^\circ+\sin45^\circ)}{\cos1^\circ.\cos2^\circ\cos3^\circ.....\cos45^\circ} = a^b$, where a & b are prime numbers and $a < b$, then a + b is equal to -

(A) 22 (B) 23 (C) 24 (D) 25

3. If $\sin\frac{\pi}{8} = \sqrt{\frac{1}{\sqrt{m}} - \frac{1}{\sqrt{n}}}$ where $m,n \in N$ then m+n is equal to-

(A) 8 (B) 6 (C) 12 (D) 10

4. A right angled trapezium is circimscribed about a circle. The radius of the circle, if the lengths of the bases (i.e. parallel sides) are equal to a and b is :

(A) a + b (B) $\frac{ab}{a+b}$ (C) $\frac{a+b}{ab}$ (D) $|a-1-b|$

5. If $\log_\alpha 8 = \gamma$, $\log_\beta \alpha = -1$ and $\log_{1/4}\beta = -1$ then $\left(\frac{1}{\alpha}+1\right)^{\log_{\sqrt{5}}(\beta^2+4\gamma^2)}$ is equal to -

(A) $\sqrt{5}$ (B) 5 (C) 25 (D) 625

6. Number of solution(s) of the equation $\log_2(x^2+3) = \frac{1}{2}\log_{1/3}\left(x+\frac{1}{x}\right)$, $x > 0$ is -

(A) 0 (B) 1 (C) 2 (D) infinite

7. For $\theta = 100^\circ$, the value of $\sin\theta + \cos\theta$ is -

(A) less than 0 (B) between 0 and 1 (C) equal to 0 (D) between 1 and $\sqrt{2}$

8. Sum of the solutions of the equation $\sin^2\theta\tan\theta + \cos^2\theta\cot\theta - \sin2\theta = 1 + \tan\theta + \cot\theta$, where $0 \le \theta \le \pi$, is -

(A) 3π (B) $\frac{3\pi}{2}$ (C) $\frac{5\pi}{2}$ (D) 7π

9. If $2^{\sec^2 x}\sqrt{\tan^2 y - \tan y + \frac{1}{2}} \le 1$, then number of values of x can be, $x \in [0,2\pi]$ -

(A) 0 (B) 1 (C) 3 (D) infinite

10. The set of values of x in $(-\pi, \pi)$ satisfying $|4\sin x - 1| < \sqrt{5}$ is given by -

(A) $\left(-\frac{9\pi}{10}, -\frac{\pi}{5}\right)$ (B) $\left(-\frac{\pi}{5}, \frac{\pi}{10}\right)$ (C) $\left(-\frac{\pi}{10}, \frac{3\pi}{10}\right)$ (D) $\left(\frac{3\pi}{10}, \frac{7\pi}{10}\right)$

Multiple Correct Answer Type (4 Marks each, –1 for wrong answer)

11. $\sqrt{\cos2x} + \sqrt{1+\sin2x} = 2\sqrt{\sin x + \cos x}$, $(n \in I)$ if -

(A) $\sin x + \cos x = 0$ (B) $x = 2n\pi$

(C) $x = n\pi - \frac{\pi}{4}$ (D) none of these

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12. The solution set of the equation $|\tan x| = \cot x$ is/are -

(A) $2n\pi + \frac{5\pi}{4}$ (B) $n\pi \pm \frac{\pi}{4}$ (C) $2n\pi + \frac{\pi}{4}$ (D) none of these

13. Values of θ satisfying the equation $(\sqrt{3}-1)\cos\theta + (\sqrt{3}+1)\sin\theta = (\sqrt{3}-1)$ is :-

(A) $2n\pi + \frac{5\pi}{6}$ (B) $2n\pi - \frac{5\pi}{6}$ (C) $2n\pi$ (D) $2n\pi \pm \frac{5\pi}{12}$

14. If $x = \sin(\alpha-\beta)\cdot\sin(\gamma-\delta)$, $y = \sin(\beta-\gamma)\cdot\sin(\alpha-\delta)$ and $z = \sin(\gamma-\alpha)\cdot\sin(\beta-\delta)$, then -

(A) $x + y + z = 0$ (B) $x + y - z = 0$ (C) $y + z - x = 0$ (D) $x^3 + y^3 + z^3 = 3xyz$

15. The equation $\log_{x+1}(x-0.5) = \log_{x-0.5}(x+1)$ has

(A) no real solution (B) no prime solution (C) an irrational solution (D) no composite solution

16. Which of the following statements does not hold good ?

(A) $\log_{10}(1.4^2 - 1)$ is positive

(B) the equation $\log_a(a+2) = 2$ is satisfied by two integral values of a

(C) $\log_{0.1}\cot\frac{3\pi}{8}$ is negative

(D) if $m = 4^{\log_4 7}$ and $n = \left(\frac{1}{9}\right)^{-2\log_3 7}$ then $n = m^4$

SECTION–III

Numerical Grid Type (Single digit Ranging from 0 to 9) (4 Marks each, –1 for wrong answer)

1. Let triangle ABC be the isosceles triangle and $\angle A = 100°$, if $\sin A + \sin B + \sin C = x$ and $\cos A + \cos B + \cos C = y$, then $(x^2 + y^2)$ is

2. If $a^x = b$, $b^y = c$, $c^z = a$ (where a,b,c, x,y,z are positive real numbers), then the value of $\frac{\sum_{r=1}^{89}\log_{10}\tan\left(\frac{\pi r}{180}\right)}{xyz}$ is equal to

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3. If $\ell n\left(\frac{a+b}{3}\right) = \left(\frac{\ell na + \ell nb}{2}\right)$, then $\frac{a}{b} + \frac{b}{a}$ is equal to -

4. Number of solution of the equation $\cos^4 x - \sin^{36} x = 1$ in $[0, 2\pi]$

5. The number of values of x lying in the interval $(-\pi, \pi)$ which satisfy the equation $81^{(1+|\sin x|+|\sin^2 x|+|\sin^3 x|+.....\infty)} = (3)^8$

6. Let $\lambda_1\lambda_2 \in [0,\pi]$ are the solutions of the equation $\operatorname{cosec}\left(\frac{\pi}{4}+x\right) + \operatorname{cosec}\left(\frac{\pi}{4}-x\right) = 2\sqrt{2}$, then $8(\sin^2\lambda_1 + \sin^2\lambda_2)$ is equal to

REVISION ASSINGMENT # 18 (QE AND SEQUENCE SERIES) MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	B	D	B	B	B	C	C	A,B,D	A,B,D
	Q.	11	12	13	14						
	A.	B,C	B,C	D	C						
SECTION-II	Q.	1	2	3	4	5	6				
	A.	5.00	1.00	14.00	469.00	9.00	5.00				
SECTION-IV	Q.1	A	B	C	D						
		P,S	P,Q,R,T	P,S	Q,S						

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